LITERATURE REVIEW ON AUTONOMOUS UNDERWATER VEHICLE (AUV) SIMULATION USING GAZEBO



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Institute for Systems Studies & Analyses (ISSA)

ISSA is involved in System Analysis, Modeling & Simulation for various defence applications pertaining to employment/deployment, tactics & force potential evaluation, tactical/strategic & mission planning etc. ISSA develops war games for all three Services.

Consequent to the reorganization of the system analysis activities within DRDO, the functions of DSE were redefined and the organization was given its present name, the Institute for Systems Studies & Analyses (ISSA), in 1980. In 1972, a small group named the Aeronautical Systems Analysis Group (ASAG) was created with the objective of carrying out aeronautical systems studies and analyses.

This group functioned from the National Aeronautics Laboratory, Bangalore, as a detachment of the Directorate of Aeronautics. In 1974, ASAG was converted into a self-accounting, full-fledged unit named the Centre for Aeronautical Systems Studies & Analyses (CASSA) and was shifted to the premises of the Aeronautical Development Establishment, Bangalore.

From 1974 to 2003, CASSA contributed significantly to a series of systems analysis studies attributed to DRDO Headquarters and the Indian Air Force, and contributed to several design and development activities and policy-level issues through its objective analyses.

In 2003, CASSA was merged with ISSA with the objective of synergizing systems analysis activities and wargaming development processes under an integrated combat environment. ISSA thereby emerged as a nodal systems analysis laboratory, and in 2013 it was placed under the SAM Cluster with a mandate in the fields of Training & Planning Wargames, Integrated Air Defence & EW, Combat Modeling, Simulation System Analysis, and computerised wargaming.

Area of Work

- Modeling, Simulation, Systems Analysis for Defence Applications
- Systems Evaluation Studies
- Combat Model Development
- Computer Science and Applications for Defence Modelling & Simulation Domain
- Operations Research and Decision Support Techniques

Vision

To transform ISSA into a centre of excellence in system analysis, modelling & simulation of defence systems to meet the challenges of present and future requirements of the armed forces.

Mission

To conduct system studies and develop high-quality integrated software for system analysis & decision support in application areas of Sensors & Weapons, Electronic Combat, Land & Naval Combat, Air-to-Air Combat and Air Defence for effective use by DRDO and the Services in Design, Mission Planning, Tactics development and Training.

DECLARATION

I hereby declare that this submission is my own work. While it includes a literature survey based on previously published works, I have appropriately acknowledged all sources within the text and references. To the best of my knowledge and belief, this work contains no material authored by another individual that has not been properly cited. Furthermore, it does not include any content that has been substantially accepted for the award of any degree or diploma from any university or other institution of higher learning, except where due acknowledgment has been appropriately made.

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Signature
Nishita Naman

CERTIFICATE

This is to certify that the Internship/Training Report titled "LITERATURE REVIEW ON AUTONOMOUS UNDERWATER VEHICLE (AUV) SIMULATION USING GAZEBO", conducted from 1/06/2026 to 31/07/2026, submitted by Nishita Naman as part of the requirements for the B.Tech degree in the Department of Information Technology at Indira Gandhi Delhi Technical University for Women, is an original record of the candidate's work carried out under my supervision.

I hereby declare that, to the best of my knowledge and belief, this submission represents the candidate's own work. Furthermore, it does not include any content that has been substantially accepted for any other degree or diploma from a university or other institute of higher learning, except where appropriate acknowledgments are provided in the text.

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Neha Aggarwal
ISSA, DRDO

Table of Abbreviations

Abbreviation	Definition
AUV	Autonomous Underwater Vehicle
CB	Center of Buoyancy
CG	Center of Gravity
DAVE	Digital / Aquatic Virtual Environment
DDS	Data Distribution Service
DOF	Degree-of-Freedom
DVL	Doppler Velocity Log
GNC	Guidance, Navigation, and Control
HIL	Hardware-in-the-Loop
IMU	Inertial Measurement Unit
LARS	Launch and Recovery System
MCM	Mine Countermeasures
ODE	Open Dynamics Engine
PeRL	Perceptual Robotics Laboratory
RL	Reinforcement Learning
ROCK	Robot Construction Kit
ROS	Robot Operating System
ROV	Remotely Operated Vehicle
SDF	Simulation Description Format
SLAM	Simultaneous Localization and Mapping
SRF	Sensor Reference Frame
URDF	Unified Robot Description Format
UUV	Unmanned Underwater Vehicle

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1. Introduction

1.1 What is an Autonomous Underwater Vehicle (AUV)

An Autonomous Underwater Vehicle (AUV) is a robotic, uncrewed submarine that drives itself through water to complete tasks without requiring real-time control from a human operator. An AUV is completely untethered, i.e., not physically bound to a surface platform, as it relies on its own onboard computers, power, and sensors to navigate and survive the extreme conditions of the deep ocean.

Unlike Remotely Operated Vehicles (ROVs), which are physically connected to a surface ship by a cable and driven by a human with a joystick, an AUV is structurally and operationally independent. AUVs are self-powered using advanced lithium-ion battery matrices and control their behaviour through onboard computational architectures and pre-programmed mission algorithms.

Since regular GPS and Wi-Fi signals attenuate within just a few metres underwater, AUVs cannot navigate the way vehicles do on land. Instead, they determine their position in 3D space by fusing data from acoustic sensors and internal motion trackers, primarily Inertial Measurement Units (IMUs) and Doppler Velocity Logs (DVLs). Eliminating the physical tether provides a significant operational advantage for marine operations: it removes cable drag, frees the robot from needing a support ship hovering directly above it, and allows it to safely explore deep or hazardous areas where a cable could become entangled.

Because of this independence, the design and validation of an AUV's guidance, navigation, and control software is considerably harder than for a tethered ROV — errors cannot be corrected in real time by a human pilot, which is precisely why simulation environments such as Gazebo have become indispensable to the development pipeline discussed later in this review.

1.2 Applications of AUVs

Because they can operate autonomously, travel long distances, and carry a wide variety of sensor payloads, AUVs have become essential tools for military, scientific, and commercial work. Their primary applications include:

- **Defense and Strategic Operations:** AUVs are crucial for Mine Countermeasures (MCM). Navies use them to safely map and identify explosive threats using advanced sonar, meaning human divers and surface ships do not have to be put in danger. They are also used to continuously gather intelligence, automatically track targets, and rapidly assess the environment in disputed coastal areas.
- **Scientific and Oceanographic Research:** In the academic world, AUVs are the go-to tool for exploring the seafloor. Researchers use them to create detailed underwater maps, survey hydrothermal vents with cameras and acoustic sensors, and collect data on water chemistry. Because they are not tied to a surface cable, they can even run missions beneath polar ice shelves — areas regular research vessels cannot reach.
- **Commercial and Industrial Infrastructure:** The offshore energy and telecom industries rely heavily on AUVs to inspect underwater structures, including subsea oil and gas pipelines, submarine internet cables, and the foundations of offshore wind farms. Using cameras and sonar, an AUV can scan miles of infrastructure to spot damage, erosion, or leaks, far faster and at lower cost than traditional cable-controlled ROVs.

Across all three domains, the common requirement is a control and perception stack that has already been proven reliable before the vehicle is ever placed in open water — which is the central motivation for simulation-based development discussed next.

1.3 Need for Simulation Before Sea Trials

Moving a robot from theoretical computer code to the actual ocean comes with major logistical, financial, and technical challenges. Unlike testing robots on land, the underwater environment is unpredictable and constantly shifting.

Running physical sea trials is expensive. It requires surface support ships, specialized equipment to launch and recover the vehicle (LARS), and a highly trained crew. On top of that, training an AUV's software to avoid obstacles, map its surroundings (SLAM), or learn to navigate typically requires thousands of trial runs.

This is where simulation becomes a vital stepping stone. By testing the robot's software in a virtual environment first, developers save considerable time and money, drastically reducing the need for expensive hardware and constant, risky field tests.

Factor	Real-World Sea Trials	Gazebo Simulation
Data Acquisition	Slow, bounded by battery life and logistics	Instantaneous, scriptable, and scannable
Edge Case Testing	Dangerous; risks the loss of a highly expensive vehicle	Safe; on collision, simply reset the environment
Ground Truth	Difficult to acquire accurately without external acoustic beacons	Absolute state parameters (x, y, z, velocity) are always known for model validation
Iteration Speed	Limited by weather windows and vessel scheduling	Can run faster than real time and in parallel
Cost per Trial	High — fuel, crew, vessel day-rate	Near-zero marginal cost per run

1.4 Why Gazebo Is Used for AUV Simulation

Gazebo has become a de facto standard for underwater robotics research because it offers a flexible, physics-based virtual world that connects directly with the Robot Operating System (ROS). For machine learning and control-systems engineers, Gazebo provides three essential features:

- **Highly Realistic Physics:** It calculates complex 3D movement (6-DOF) to recreate real-world physical forces, simulating how water resistance, buoyancy, and ocean currents push and pull on the robot.
- **Realistic Sensor Noise:** Real-world marine environments are subject to severe acoustic and optical disturbances. Gazebo lets developers simulate fading light, murky water, and sound interference, which is crucial for adding realistic "noise" to training data so that computer-vision and sensor-fusion algorithms learn to handle imperfect, real-world conditions rather than only "perfect" simulated data.
- **Seamless Code Deployment:** Because Gazebo is fully integrated with ROS, developers do not have to change their code when moving from simulation to the real world. A Python script for machine learning or a C++ program for mapping (SLAM) will interact with the virtual robot the same way it would interact with the actual physical hardware.

This close coupling between simulated physics, simulated sensing, and unmodified control code is what makes Gazebo attractive not only to individual researchers but also to defence and industrial organisations that need a reproducible, auditable path from algorithm design to sea trial — a theme that recurs throughout the literature reviewed in Section 5.

2. Overview of the Gazebo Simulator

2.1 Gazebo Architecture

Gazebo operates on a decoupled (separated) client-server architecture. It is divided cleanly into a computational backend (gzserver) and a graphical frontend (gzclient). The gzserver handles the core execution loop, including physics computation, sensor generation, and state updates — effectively the "brain" of the simulator — while gzclient handles 3D rendering and the user interface via the Object-Oriented Graphics Rendering Engine (OGRE). Because these two parts are split, they do not always have to run together. This allows researchers to run computationally intensive simulations, such as reinforcement-learning training loops or multi-robot swarm missions, in a "headless" mode without rendering graphics, saving significant CPU and GPU overhead.

Gazebo builds its virtual marine worlds and physical robot descriptions using a system called the Simulation Description Format (SDF). While another format called URDF is also common in robotics, SDF has several upgrades that make it the better choice for underwater vehicles:

- **Nested Models:** Complex robotic assemblies can be grouped inside one another.
- **Advanced Friction:** Finer, more realistic control over how surfaces interact.
- **Custom Plugins:** Developers can easily attach their own code to simulate specific, real-world behaviours.

Because of these features, SDF is well suited to handling the complicated hull geometries and hydrodynamic effects required to simulate underwater robots.

2.2 Physics Engines

Gazebo provides modular support for multiple physics engines, including the Open Dynamics Engine (ODE), Bullet, DART, and Simbody. By default, ODE manages rigid-body dynamics and contact collisions. However, standard physics engines do not natively calculate fluid-structure interactions specific to underwater environments. To address this, marine robotics researchers have developed specialized simulation packages that sit on top of the core engine:

- **The uuv_simulator Package:** Developed by Manhães et al. (2016), this package utilizes custom Gazebo plugins to calculate hydrodynamic forces based on Fossen's equations of marine craft motion. It treats fluid forces as external disturbances applied to rigid-body links.
- **The freefloating_gazebo Package:** In contrast to the link-by-link plugin forces of uuv_simulator, freefloating_gazebo alters fluid-dynamics modelling at the foundational engine level. It computes buoyancy and fluid resistance directly from the geometry and submerged volume of the vehicle's links, making it highly effective for simulating underwater structures that cross the air-water interface or exhibit irregular surface geometries.

The existence of two philosophically different approaches — plugin-based force injection versus geometry-driven computation at the engine level — is itself a useful lens for reading the literature survey in Section 5: successive papers can generally be classified as extending one approach, bridging the two, or replacing them with GPU-accelerated alternatives.

2.3 Sensor Simulation Capabilities

Simulating an AUV requires accurate modelling of aquatic sensor payloads, which are susceptible to signal degradation from acoustic and visual disturbances (Lakhyari et al., 2024). Through frameworks such as the DAVE Aquatic Virtual Environment project, Gazebo replicates specialized marine instruments (Zhang et al., 2022):

- **Doppler Velocity Logs (DVLs):** Simulated by casting ray vectors from the AUV model to the seafloor, measuring Doppler frequency shifts to compute ground-relative 3D velocity vectors.
- **Acoustic Sonars:** Packages like DAVE utilize GPU-accelerated ray-tracing to simulate multibeam, forward-looking, and side-scan sonars, generating realistic raw acoustic backscatter data and point clouds.
- **Inertial Measurement Units (IMUs):** Simulates high-frequency angular velocities and linear accelerations, complete with configurable bias drift and Gaussian noise models to replicate real-world physical errors.

Gazebo supports structural noise injection directly within its sensor plugins. For visual sensors, developers can configure shaders to simulate water turbidity, light absorption, and scattering effects. For acoustic instruments, high-frequency stochastic noise models are injected into the data streams, ensuring downstream computer-vision and sensor-fusion algorithms are resilient to the imperfections of real-world underwater environments.

2.4 ROS/Gazebo Integration

2.4.1 ROS 1: The Original Middleware and the Publish/Subscribe Architecture

The interplay between Gazebo and the Robot Operating System (ROS) relies on a standardized, asynchronous publish/subscribe architecture. Simulated sensors attached to the AUV model generate data packages within Gazebo and publish them to specific ROS topics. Conversely, the AUV's autonomous control nodes subscribe to these topics to ingest state data, execute control algorithms, and publish calculated thrust vectors back to actuator topics.

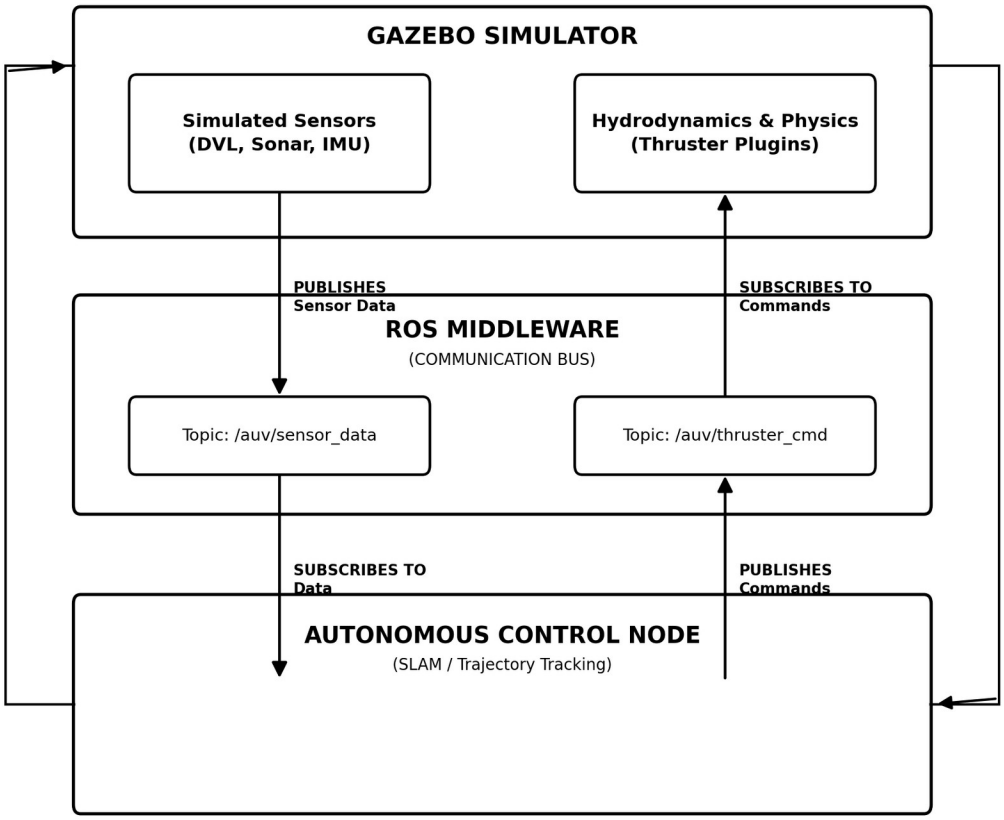


Figure 1: Publish/Subscribe data flow between the Gazebo Simulator, the ROS middleware, and the Autonomous Control Node.

The first generation of this integration, built on ROS 1 (the version used by the original uuv_simulator package described in Section 5.5), rests on a small set of core concepts that every Gazebo-based AUV project of that era depends on:

- **roscore / Master Node:** A single, centralized process that acts as a name server, allowing every other node in the system — simulated sensor drivers, controllers, planners — to find and connect to one another. Every node must register with this master at startup.
- **Topics (Publish/Subscribe):** The asynchronous, many-to-many messaging pattern shown in Figure 1 above, used for continuous data streams such as sensor readings and thruster commands.
- **Services (Request/Response):** A synchronous, one-to-one call-and-response pattern used for occasional operations, such as resetting the simulated world or querying a vehicle's current state on demand.
- **TF (Transform Tree):** A library that tracks the relationship between multiple coordinate frames over time — for example, the transform between the Body-Fixed Frame and the Inertial Frame introduced in Section 3.1 — so that sensor data expressed in one frame can be automatically converted into another.
- **catkin Build System and rosbag:** catkin compiles the interdependent C++/Python packages that make up a typical uuv_simulator project, while rosbag records and replays the full stream of simulated topics for later offline analysis — an essential tool for debugging a control algorithm after a failed simulated mission.

This architecture worked well for single-vehicle, single-institution research and is exactly what Watanabe et al. (2015) and Manhães et al. (2016) built upon in Section 5. However, as the literature survey traces through 2017-2022, the ROS 1 architecture increasingly became a limiting factor rather than an enabler, for several concrete reasons:

- **Single Point of Failure:** Because every node depends on the central roscore master to discover its peers, the failure or restart of that one process brings down the entire simulated system — a serious liability when trying to run long, unattended reinforcement-learning training loops (Section 5.6) or multi-vehicle swarm missions (Section 5.7).
- **No Native Real-Time Guarantees:** ROS 1's underlying transport (TCPROS) was not designed for deterministic, bounded-latency delivery, which is a problem for Hardware-in-the-Loop (HIL) setups where real AUV flight-control hardware expects a predictable, jitter-free stream of simulated sensor data.
- **Poor Multi-Robot / Multi-Vendor Scalability:** Every robot in a ROS 1 system typically shares (or bridges into) the same master and topic namespace. As papers such as Sampathkumar et al. (2022) began simulating fleets of AUVs communicating over bandwidth-limited acoustic links, this single shared namespace became increasingly awkward to partition, secure, and scale.
- **Security Model:** ROS 1 topics and services have no built-in authentication or encryption, which is acceptable for a closed academic testbed but a genuine concern for defence-relevant simulation environments where multiple organisations or classification levels might eventually need to interoperate.
- **End-of-Life Timeline:** The final ROS 1 long-term-support release (Noetic Ninjemys) reaches end-of-life alongside its underlying Ubuntu 20.04 platform, meaning ROS 1-based marine simulation stacks face an approaching point at which they will no longer receive security patches or compatible dependencies.

2.4.2 Why the Field Moved to ROS 2, and What It Changed

The marine robotics field is undergoing a systematic migration from ROS 1 to ROS 2, which utilizes a decentralized Data Distribution Service (DDS) communication layer. While older architectures like the original uuv_simulator relied heavily on the master-node communication style of ROS 1, modern simulations utilize Project Plankton —

the open-source port of uuv_simulator built specifically for ROS 2 and the modernized Ignition/Gazebo Harmonic ecosystem. This transition introduces real-time processing guarantees, deterministic node execution, and optimized multi-robot network topologies, which are critical for simulating synchronized multi-AUV swarms.

Concretely, ROS 2 replaces the single-master architecture with a set of features that address each of the ROS 1 limitations listed above:

- **Decentralized Discovery via DDS:** Nodes find and talk to one another directly through a Data Distribution Service layer, with no central master process required. Any single node — including a simulated AUV or a ground-station monitor — can join or leave the network without taking the rest of the system down, which is essential for the multi-vehicle swarm scenarios discussed in Sections 5.7 and 5.8.
- **Configurable Quality-of-Service (QoS):** Every topic in ROS 2 can be tuned with QoS policies — reliability (best-effort vs. guaranteed delivery), durability (whether late-joining nodes receive the last published message), and deadline/liveliness settings — allowing a designer to trade determinism against bandwidth in exactly the way a bandwidth-limited underwater acoustic link (Sampathkumar et al., 2022) requires.
- **Real-Time and Deterministic Execution:** ROS 2's executor model and support for real-time-safe DDS implementations give the platform the deterministic, bounded-latency behaviour that Hardware-in-the-Loop AUV validation (Watanabe et al., 2015) was straining to achieve on top of ROS 1.
- **Native Multi-Robot Namespacing:** Because there is no shared master, each simulated AUV in a swarm can run its own independent graph of nodes, cleanly namespaced, and still discover fleet-mates only where explicitly configured — directly supporting the decentralized, peer-to-peer swarm communication highlighted in Section 5.8.
- **Security (SROS2):** ROS 2 adds an optional security layer (SROS2) built on DDS-Security, providing node authentication, access control, and encrypted communication — closing the security gap identified as a ROS 1 limitation above.
- **Actions and Improved Lifecycle Management:** ROS 2 formalizes long-running, preemptable, goal-based tasks (such as "navigate to waypoint" or "complete survey pattern") as first-class Actions with feedback and cancellation, and introduces managed node lifecycles (unconfigured → inactive → active → finalized) that make it easier to bring up, reconfigure, or safely shut down a complex AUV software stack without leaving it in an undefined state.
- **colcon Build System and ros_gz Bridge:** catkin is replaced by colcon, and the direct Gazebo-classic plugin interface used by the original uuv_simulator is replaced by the ros_gz bridge, which connects ROS 2 topics to the modernized Gazebo (Ignition/Harmonic) transport layer — the specific piece of plumbing that Project Plankton (Section 5.8) had to rebuild when porting uuv_simulator forward.

Taken together, the ROS 1-to-ROS 2 migration is not a cosmetic version bump: it is a direct, feature-for-feature response to the operational limitations — single points of failure, lack of real-time guarantees, weak multi-robot scaling, and absent security — that a decade of ROS 1-based AUV simulation research had exposed. This is precisely the same "gap identified, gap closed" pattern described in Section 5.9, applied to the underlying middleware itself rather than to the physics or perception layers above it.

2.5 Advantages and Limitations

Factor	Gazebo Simulation Advantages	Gazebo Simulation Limitations
Ecosystem	Compatible with ROS/ROS 2; extensive library of marine assets	Dependency on middleware version matching; complex installation configurations
Physics	Handles 6-DOF rigid-body kinetics and hydrodynamic overlays	Hydrodynamic coefficients are idealized; struggles with transient fluid phenomena (e.g., turbulent wakes)
Validation	Zero-risk environment for training RL loops and testing SLAM algorithms	"Sim-to-real" gap; perception models often fail under non-uniform real-world turbidity
Efficiency	Headless deployment (gzserver) enables parallel computing	High-fidelity ray-tracing for acoustic payloads creates computational bottlenecks

3. AUV Dynamic Modelling & Related Information

3.1 Coordinate Systems and 6-DOF Model

To accurately track and control a submerged tactical asset, the navigation software must continuously translate data between two distinct mathematical reference frames. Understanding how these frames interact is the foundation of the vehicle's entire Guidance, Navigation, and Control (GNC) stack.

- **Inertial / Earth-Fixed Frame (XE, YE, ZE):** This is the static, global map. The origin is usually pinned to a fixed geographic location, such as the initial deployment coordinates from a naval vessel or a harbor dock. Engineers use this stationary frame to define mission waypoints, plot the overall trajectory, and measure the vehicle's absolute positional drift over time.
- **Body-Fixed Frame (XB, YB, ZB):** This is a dynamic, moving coordinate system physically attached to the AUV itself, with its origin placed directly at the vehicle's Center of Gravity (CG). This frame is critical for the flight software because onboard sensors like IMUs and DVLs measure data relative to the vehicle's current orientation. Furthermore, all thruster forces and control-surface deflections are computed and applied within this localized frame.

The transformation between these frames defines the vehicle's 6-Degree-of-Freedom (6-DOF) state, encompassing three translational movements (surge, sway, heave) and three rotational movements (roll, pitch, yaw).

3.2 Rigid-Body Equations of Motion

Gazebo simulation physics incorporates the non-linear Newton-Euler equations of motion. AUVs require a full dynamic formulation to account for hydrodynamic coupling, generally written in the compact form:

$$M_{RB} \cdot \dot{\nu} + C_{RB}(\nu) \cdot \nu = \tau_{actuator} + \tau_{hydrodynamic}$$

Breaking this down into its practical engineering components:

- **M_{RB} (Rigid-Body Mass and Inertia Matrix):** Accounts for the physical dry weight of the vehicle and how its mass is distributed across the hull geometry.
- **$C_{RB}(\nu)$ (Coriolis and Centripetal Matrix):** As the AUV rotates — especially during aggressive tactical manoeuvres like sharp yaw turns — this matrix calculates the fictitious forces acting on the vehicle due to the rapid rotation of the body-fixed frame.
- **ν and $\dot{\nu}$ (Velocity and Acceleration):** These vectors represent the vehicle's linear and angular speeds, as well as their rate of change over time.
- **$\tau_{actuator}$ (Control Inputs):** The active mechanical forces and torques commanded by the autopilot, such as propeller thrust or the mechanical deflection of steering rudders.
- **$\tau_{hydrodynamic}$ (External Environment):** Captures how the ocean pushes back — the drag, buoyancy, and added-mass forces exerted by the surrounding water, which fundamentally alter how the vehicle handles and accelerates.

This formulation, first systematized for marine craft by Fossen (1994), is the mathematical backbone that later Gazebo-based tools — most notably the uuv_simulator package discussed in Section 5.5 — implemented directly as executable plugin code, closing the gap between naval architecture theory and real-time simulation.

4. Hydrodynamic Modelling in Gazebo

4.1 Added Mass Matrix

Because water is a dense, incompressible fluid medium, an accelerating AUV must physically push away and displace a volume of water equal to its own directional motion. This fluid resistance creates an inertial lag known as "added mass." In Gazebo, this is implemented by introducing a non-diagonal 6×6 added-mass matrix, MA, that scales with the hull geometry. This is mathematically combined with the rigid-body mass matrix to dictate true acceleration properties:

$$MTotal = MRB + MA$$

4.2 Hydrodynamic Damping

Hydrodynamic damping models energy loss due to skin friction and vortex shedding. The Gazebo plugin calculates drag by combining linear damping matrices (dominant at low speeds) and quadratic drag tensors (dominant during high-speed manoeuvres). These forces are computed dynamically based on the vehicle's velocity relative to environmental ocean currents.

4.3 Restoring Forces (Buoyancy and Gravity)

A defence-grade AUV cannot rely purely on software and thrusters to stay upright; it must be inherently stable in the water even if power fails. This submerged stability is dictated by hydrostatic restoring forces, which rely on the continuous interplay between two fundamental physical properties:

- **Gravity (W = mg):** The total physical weight of the vehicle, pulling directly downward through its Center of Gravity (CG).
- **Buoyancy (B = ρg∇):** The upward lift exerted by the displaced water, acting through the vehicle's geometric Center of Buoyancy (CB) in accordance with Archimedes' principle, where ρ represents water density and ∇ represents the displaced volume.

For stability, naval engineers ensure the Center of Buoyancy is positioned vertically above the Center of Gravity. This structural offset generates a metacentric restoring torque, which automatically returns the vehicle to a level orientation following external disturbances, serving as a passive safety mechanism — functioning like a heavy pendulum that swings the vehicle back to a stable, level orientation without requiring a single line of code, thruster input, or battery power.

Together, Sections 3 and 4 define the complete dynamic and hydrostatic model that any credible AUV-capable simulator must reproduce. As will become clear in the literature survey that follows, the single most important historical event in the Gazebo ecosystem was the point at which these equations were, for the first time, implemented as native, reusable Gazebo plugins rather than external, hand-built approximations.

5. Literature Survey: The Chronological Evolution of Gazebo in AUV Simulation

The transition of Autonomous Underwater Vehicles from theoretical mathematical models to reliable sea trials has been heavily dependent on the evolution of physics-based simulation middleware. Read strictly in publication order, the literature demonstrates a clear trajectory: a general-purpose terrestrial simulator is created, the marine-robotics community identifies its shortcomings, middleware bridges are built to prove real-time feasibility, a dedicated hydrodynamics package finally closes the physics gap, and subsequent work layers machine learning, high-fidelity sensing, and multi-vehicle coordination on top of that foundation. Section 5.1 presents this chronology as a single reference table; Sections 5.2 to 5.8 then walk through each period in the depth the question warrants, answering directly when the very first paper in this lineage was published and what it established.

5.1 Chronological Timeline at a Glance

Year	Author(s)	Publication	Core Contribution
2004	Koenig & Howard	Design and use paradigms for Gazebo, an open-source multi-robot simulator (IEEE/RSJ IROS)	Introduces Gazebo itself as a 3D, multi-robot, rigid-body simulator (terrestrial focus)
2009	Brown, Kim & Eustice	Overview of AUV research and testbed at PeRL (Tech. Report)	Documents the lack of standardized underwater simulation tooling prior to Gazebo's marine adoption
2015	Watanabe et al.	The Rock-Gazebo integration and a real-time AUV simulation (LARS/SBR)	Bridges Gazebo with the Rock framework; proves real-time Hardware-in-the-Loop feasibility
2016	Manhães et al.	UUV Simulator: A Gazebo-based package for underwater intervention and multi-robot simulation (IEEE/MTS OCEANS)	Injects Fossen's hydrodynamic equations natively into Gazebo via C++ plugins — the paradigm shift
2020	Q. Zhang, Lin, Sha & Li	Deep interactive reinforcement learning for path following of AUV (IEEE Access)	Uses matured Gazebo hydrodynamics to train RL-based path-following controllers
2022	Mabel Zhang et al.	DAVE Aquatic Virtual Environment: Toward a general underwater robotics simulator (IEEE/MTS OCEANS)	Adds GPU-accelerated sonar/DVL ray-tracing and turbidity shaders — closes the perception gap
2022	Sampathkumar et al.	Plume tracing simulations using multiple AUVs (WUWNet)	Extends Gazebo to multi-vehicle, bandwidth-limited swarm coordination scenarios

Year	Author(s)	Publication	Core Contribution
2023	Sang & Norris	An AUV simulation with fuzzy sensor fusion for pipeline inspection (IEEE Sensors Journal)	Demonstrates industrial-grade inspection workflows on ROS 2-era Gazebo
2023-Present	Community (Project Plankton)	ROS 2 port of uuv_simulator for Ignition/Gazebo Harmonic	Migrates the ecosystem to DDS-based, deterministic, swarm-ready middleware
2024	Lakhyari, Boulaala, Mekrini & El Mrabet	AUV: An overview of current technologies and applications (ICISEEC'24)	Contextualizes sensor degradation challenges that motivate high-fidelity simulated sensing

5.2 2004: The Origin — Koenig and Howard Introduce Gazebo

The very first paper in this lineage, and the one that makes everything else in this review possible, was published in 2004: Nathan Koenig and Andrew Howard's "Design and use paradigms for Gazebo, an open-source multi-robot simulator," presented at the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). This is the foundational publication that introduced Gazebo to the robotics community — more than a decade before any AUV-specific extension existed.

What actually happened in 2004 was not the creation of an underwater simulator at all. Koenig and Howard's stated objective was to build a 3D, dynamic, multi-robot environment with high-fidelity rigid-body physics, aimed primarily at outdoor terrestrial and planetary-rover research. Gazebo's underlying physics architecture was built on the Open Dynamics Engine (ODE), which made it strictly optimized for gravity, surface friction, and rigid-body collisions — the concerns of a wheeled or legged robot moving across dry or planetary terrain, not a vehicle submerged in a dense, incompressible fluid.

The significance of this starting point for the present review is twofold. First, it explains why, for more than a decade afterward, Gazebo simply had no concept of buoyancy, added mass, or hydrodynamic drag — these are absent from the original 2004 design because they were never part of the problem the authors were solving. Second, it establishes the client-server architecture (gzserver/gzclient) and the plugin-based extensibility model described in Section 2.1, both of which remained structurally unchanged even as later authors bolted marine physics onto the platform. Every subsequent milestone in this survey is, in effect, the story of the marine-robotics community repurposing a 2004 terrestrial simulator for an environment its original authors never designed it to handle.

5.3 2004-2010: The Terrestrial Origins and Kinematic Limitations

During the years immediately following Gazebo's release, marine robotics research proceeded largely in parallel with, rather than inside, the Gazebo ecosystem. Brown, Kim, and Eustice (2009), documenting the AUV testbeds at the University of Michigan's Perceptual Robotics Laboratory (PeRL), illustrate the state of the field at this time and highlight a massive bottleneck: researchers lacked standardized fluid-structure interaction models.

Without a shared, general-purpose underwater physics engine, laboratories were forced to choose between two unsatisfying alternatives: (i) rely on expensive, logistically heavy physical sea trials for every iteration of a control algorithm, or (ii) build highly simplified, proprietary kinematic simulators that could not accurately replicate the non-linear fluid dynamics of the ocean — no added mass, no realistic damping, and no coupled six-degree-of-freedom motion. The absence of native buoyancy, added-mass, and hydrodynamic-drag calculations in Gazebo

meant that, prior to roughly 2015, the platform was largely unsuitable for serious AUV development, despite already being popular for terrestrial robotics.

This period is best understood as a waiting stage: the general-purpose tooling (Gazebo) existed, the mathematical theory of marine-craft dynamics (Fossen's equations, formalized in the 1994 text referenced in Section 3.2) existed, but no one had yet written the software bridge connecting the two inside an open, reusable simulator. That bridge would begin to appear in the next period.

5.4 2011-2015: Middleware Integration and Hardware-in-the-Loop (HIL)

A major milestone toward standardized marine simulation occurred when researchers began forcing generic physics engines to interact with robust robotics middleware in order to achieve real-time, deterministic execution. Watanabe et al. (2015) published "The Rock-Gazebo integration and a real-time AUV simulation," presented at the Latin American Robotics Symposium (LARS) and the Brazilian Symposium on Robotics (SBR), demonstrating how to bridge Gazebo with the Robot Construction Kit (Rock) framework.

This research mattered because it proved, for the first time in the open literature, that Gazebo could satisfy the strict timing constraints required for Hardware-in-the-Loop (HIL) validation of AUV architectures. By establishing a synchronous tick-rate between the software controller and the physics engine, Watanabe et al. achieved real-time simulation parity — meaning simulated time and wall-clock time advanced together closely enough that real AUV control hardware could be connected directly to the simulation loop and behave as it would in the field.

However, the literature of this period candidly acknowledges significant remaining gaps. Visual rendering of the underwater environment and fluid damping were still not natively supported inside Gazebo itself. Researchers had to heavily customize the simulator or rely on external graphical libraries such as osgOcean to approximate light refraction and water surface physics. In other words, by 2015 the field had solved the timing/middleware problem but not yet the physics problem — the two would finally be solved together in the very next year.

5.5 2016: The UUV Simulator Breakthrough (The Paradigm Shift)

The definitive turning point in the literature is the 2016 publication of "UUV Simulator: A Gazebo-based Package for Underwater Intervention and Multi-Robot Simulation" by Manhães, Scherer, Voss, Douat, and Rauschenbach, presented at IEEE/MTS OCEANS in Monterey, California. This paper revolutionized the field by directly injecting Fossen's equations of marine-craft motion into the Gazebo/ODE physics loop through custom C++ plugins, rather than treating hydrodynamics as an external add-on layered awkwardly on top of the simulator.

Prior to this paper, hydrodynamics were treated as external, often inaccurate mathematical estimates bolted onto a terrestrial engine. Manhães et al. instead created a modular architecture that formally introduced two reusable components that essentially every later Gazebo-based AUV paper builds upon:

- **The Hydrodynamic Plugin:** Allowing Gazebo to compute complex 6-DOF non-linear equations, including added-mass matrices (the virtual mass of water displaced by the AUV's acceleration), Coriolis forces, and quadratic hydrodynamic damping (skin friction and vortex shedding).
- **The Hydrostatic Plugin:** Introducing inherent restoring forces by actively computing buoyancy based on the AUV's dynamically calculated displaced volume and the relationship between its Center of Gravity (CG) and Center of Buoyancy (CB).

The consequence of this single paper is difficult to overstate: the open-source uuv_simulator package it produced established ROS 1 plus Gazebo as the de facto academic standard for underwater simulation, eliminating the need for individual research teams to build custom physics engines from scratch. Every subsequent development discussed below — reinforcement learning, GPU-accelerated sonar, multi-vehicle swarms, and the ROS 2 migration — is, in a very direct sense, an extension of the plugin architecture Manhães et al. introduced in 2016.

5.6 2017-2020: The Rise of Machine Learning and Advanced Control

With foundational hydrodynamics standardized via uuv_simulator, the focus of academic papers shifted toward leveraging Gazebo's computational efficiency to train artificial intelligence. Deep learning and reinforcement learning (RL) require millions of trial-and-error iterations, a task that is financially and physically impossible to execute in the actual ocean, but became routine once a validated, headless, fast-forwardable physics simulation was available.

Q. Zhang, Lin, Sha, and Li (2020) published "Deep interactive reinforcement learning for path following of autonomous underwater vehicle" in IEEE Access, marking a significant application of Gazebo in AI-driven control. They utilized the simulator's now-matured hydrodynamic models to safely execute thousands of automated training epochs. Their work demonstrated that control algorithms trained purely in Gazebo's simulated fluid environment could successfully learn complex, non-linear path-following behaviours robust enough to handle simulated current disturbances — a result that would have been prohibitively expensive, and arguably unsafe, to obtain through repeated physical sea trials.

This period reframed Gazebo's role in the research pipeline: it was no longer merely a tool for verifying that a hand-designed controller worked, but a training ground in which the controller itself could be discovered automatically. That shift set the stage for the next bottleneck the community would need to solve — not control, but perception.

5.7 2021-2022: Closing the Perception Gap with Project DAVE

As autonomous control software matured, a new bottleneck emerged: sensor perception. Real-world marine environments are characterized by severe acoustic noise and optical degradation, phenomena that earlier Gazebo-based models largely failed to reproduce, since the 2016 hydrodynamic plugins had focused on forces and motion rather than on sensing fidelity.

To bridge this "sim-to-real" perception gap, Mabel Zhang et al. (2022) published "DAVE Aquatic Virtual Environment: Toward a General Underwater Robotics Simulator" at IEEE/MTS OCEANS in Hampton Roads, Virginia. Project DAVE built upon uuv_simulator but focused intensely on sensor fidelity, introducing GPU-accelerated ray-tracing specifically designed to model acoustic payloads. Through DAVE, Gazebo gained the ability to:

- **Simulate Doppler Velocity Logs (DVLs)** by casting ray vectors to a simulated bathymetric seafloor to calculate Doppler shifts and acoustic bottom-tracking.
- **Render high-fidelity Multibeam Sonar point clouds,** complete with stochastic Gaussian noise and acoustic backscatter.
- **Introduce visual shaders for water turbidity,** forcing computer-vision algorithms to operate in murky, light-absorbed environments rather than idealized, perfectly clear simulated water.

Simultaneously, the literature of this period began addressing multi-vehicle operations. Sampathkumar et al. (2022) published "Plume tracing simulations using multiple autonomous underwater vehicles" at the International Conference on Underwater Networks & Systems (WUWNet), utilizing Gazebo to simulate a fleet executing cooperative environmental surveys. Their work highlighted the critical necessity of simulating acoustic communication limitations — such as low bandwidth and signal attenuation — between swarm agents, a constraint that single-vehicle papers up to this point had generally ignored. Read together, the 2021-2022 period essentially split into two complementary threads: making one vehicle perceive its environment realistically (DAVE), and making many vehicles coordinate realistically despite a hostile communication channel (Sampathkumar et al.).

5.8 2023-Present: Multi-Agent Swarms, ROS 2, and Digital Twins

Modern literature reflects a systematic architectural migration away from ROS 1, driven by the need to simulate massive AUV swarms without encountering the bottleneck of a centralized master node. The field is currently dominated by the transition to ROS 2, which uses a Data Distribution Service (DDS) communication layer, via initiatives like Project Plankton, which ports the legacy uuv_simulator into the modern ROS 2 and Gazebo Harmonic (Ignition) ecosystem. This decentralized architecture provides deterministic, real-time processing guarantees necessary for peer-to-peer swarm communication — directly addressing the bandwidth and coordination concerns that Sampathkumar et al. (2022) had identified as unresolved just a year earlier.

Recent papers, such as Sang and Norris (2023), emphasize the practical, industrial application of these environments, demonstrating Gazebo's use in developing fuzzy sensor-fusion techniques for autonomous underwater pipeline inspection in IEEE Sensors Journal — showing that the platform has moved well beyond pure academic demonstration into workflows resembling deployable commercial inspection systems. Furthermore, cutting-edge research continues to explore alternative physics implementations, such as the freefloating_gazebo package described in Section 2.2, which computes buoyancy directly from the geometric mesh of submerged links, proving highly effective for simulating AUVs crossing the turbulent air-water interface — a scenario the original 2016 uuv_simulator plugins were not specifically designed to handle.

Taken together, this most recent era represents less a single breakthrough (as in 2016) and more a broadening of the ecosystem along three simultaneous axes: middleware modernization (ROS 2/DDS), physics diversification (freefloating_gazebo alongside uuv_simulator), and application maturity (industrial pipeline inspection alongside academic swarm research).

5.9 Cross-Cutting Observations on the Evolution

Reading Sections 5.2 through 5.8 end-to-end, three broad phases stand out. From 2004 to roughly 2010, Gazebo existed purely as a terrestrial tool, and the marine-robotics community had no shared simulation substrate at all. From 2011 to 2016, the community first solved the middleware/timing problem (Watanabe et al., 2015) and then, decisively, the physics problem (Manhães et al., 2016), together transforming Gazebo from an unsuitable terrestrial engine into the field's standard AUV simulator. From 2017 onward, with the core physics essentially settled, research energy shifted upward in the software stack — first to learning-based control (2017-2020), then to high-fidelity perception (2021-2022), and finally to multi-agent coordination and next-generation middleware (2023-present).

A second observation is that almost every major step in this history was a response to a previously identified gap. The 2009 PeRL testbed report identified the absence of fluid-structure modelling; the 2016 uuv_simulator paper closed exactly that gap. The 2015 Rock-Gazebo paper noted that visual rendering and fluid damping were still unsupported; DAVE (2022) closed the sensing half of that gap, while the 2016 hydrodynamic and hydrostatic plugins had already closed the damping half. Sampathkumar et al. (2022) flagged unresolved bandwidth and coordination limits for swarms; the ROS 2/DDS migration and Project Plankton directly target that limitation today. This pattern — gap identified, gap closed by the next major publication — is arguably the most useful single insight for anyone planning new work in this space: the next paper is usually written to fix the last paper's stated limitation.

6. Conclusion

6.1 Summary of AUV Simulation Benefits

Developing and deploying Autonomous Underwater Vehicles is a complex, risky, and expensive process. Simulation has proven to be an indispensable step in modern robotics projects, as it saves significant time and resources during the development phase by reducing hardware requirements and the need for frequent, expensive field-test deployments. Through simulation frameworks like Gazebo, researchers can safely test algorithmic responses to harsh underwater environments without risking the loss of physical prototypes.

Gazebo's modular architecture, and specifically its ability to integrate with prominent robotics frameworks such as ROS and Rock, provides a powerful tool for Hardware-in-the-Loop (HIL) simulations. By allowing developers to accurately model 6-DOF rigid-body equations, hydrodynamic damping, and complex sensor payloads like Doppler Velocity Logs and multibeam sonars, Gazebo empowers engineers to validate guidance, navigation, and control systems before a vehicle ever touches the water.

As the chronological survey in Section 5 shows, this capability was not present from the outset. It is the cumulative result of two decades of incremental, gap-driven contributions: a terrestrial simulator released in 2004, a middleware bridge proven in 2015, a native hydrodynamics package delivered in 2016, and a decade of subsequent refinement in learning, perception, and swarm coordination.

6.2 Future Scope

While current simulation environments provide robust physics and kinematic testing, the trajectory of AUV research points toward several areas requiring advanced simulation capabilities:

- **More Realistic Visuals:** As AUVs rely increasingly on cameras for navigation, simulators are needed that can generate photorealistic underwater environments out of the box. Future tools should natively handle tricky effects like water turbidity, light refraction, and scattering without depending on extra graphical plugins.
- **Better Coordination for Multi-Vehicle Systems:** As interest grows in using AUV swarms for wide-area surveys, simulators need to better model the harsh reality of underwater communication — specifically low bandwidth and high signal attenuation. This is crucial for accurately testing how teams of robots share data and coordinate in real-world scenarios.
- **Support for Soft and Biomimetic Designs:** AUVs are beginning to take inspiration from marine life to improve manoeuvrability and energy efficiency. Future simulation frameworks must move beyond standard rigid-body mechanics to accurately test soft actuators and biomimetic fin propulsion.
- **Advanced AI and Sensor Fusion Training:** Because deep learning and sensor fusion are becoming essential for navigating complex, unstructured environments, simulators will serve as the primary training ground. Future systems must be optimized to handle the heavy computational load required to train these advanced AI models.
- **Consolidation Around ROS 2 Digital Twins:** Given the trajectory identified in Section 5.8, it is reasonable to expect that ongoing efforts such as Project Plankton will continue to mature into full digital-twin pipelines, where a single Gazebo/ROS 2 description of an AUV is used simultaneously for design, control-software training, and live monitoring during actual missions.

7. References

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